



Machine Learning

Lecture 4: Regression

Asst. Prof. Dr. Santitham Prom-on

Department of Computer Engineering, Faculty of Engineering
King Mongkut's University of Technology Thonburi



Lecture 3 - Topics

- Linear regression, constraints
- Poisson regression
- Survival analysis

Lab

https://drive.google.com/file/d/12lm4MO-xYqFZb-gp_VChhDncPjg_PPet/view?usp=sharing

Simple linear regression

- Simple linear regression model has one input x and one output y .
- The relationship can be explain as the following equation

$$f(x) = w_0 + w_1x$$

- Mathematically, parameters are obtained by least square method

Multiple regression

- Multiple linear regression model structure is exactly the same as the linear regression

$$f(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + \dots$$

- Mathematically, parameters are obtained by least square method

Multiple regression with interaction

- Adding interaction terms to a regression model can greatly expand understanding of the relationships among the variables in the model
- This occurs when two or more variables depend on one another for the outcome

$$f(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + w_3x_1x_2 + \dots$$

Feature selection via shrinkage (Regularization)

- Instead of explicitly selecting features, in some approaches we can bias the learning process towards using a small number of features
- Key ideas: objective function has two parts
 - Term representing error minimization
 - Term that shrinks parameters toward 0



Shrinkage and regression

- Consider the case of linear regression

$$y = f(x) = w_0 + \sum_{i=1}^n w_i x_i$$

- The standard approach minimizes sum squared error

$$\begin{aligned} E(w) &= \sum_{d \in D} \left(y^{(d)} - f(x^{(d)}) \right)^2 \\ &= \sum_{d \in D} \left(y^{(d)} - w_0 - \sum_{i=1}^n w_i x_i^{(d)} \right)^2 \end{aligned}$$

Ridge regression and the Lasso

- Ridge regression adds a penalty term, the L2 norm of the weights

$$E(w) = \sum_{d \in D} \left(y^{(d)} - w_0 - \sum_{i=1}^n w_i x_i^{(d)} \right)^2 + \lambda \sum_{i=1}^n w_i^2$$

- The Lasso method adds a penalty term, the L₁ norm of the weights

$$E(w) = \sum_{d \in D} \left(y^{(d)} - w_0 - \sum_{i=1}^n w_i x_i^{(d)} \right)^2 + \lambda \sum_{i=1}^n |w_i|$$



Lasso optimization

- Lasso stands for “least absolute shrinkage and selection operator

$$E(w) = \sum_{d \in D} \left(y^{(d)} - w_0 - \sum_{i=1}^n w_i x_i^{(d)} \right)^2 + \lambda \sum_{i=1}^n |w_i|$$

- This is equivalent to the following constrained optimization problem

$$E(w) = \sum_{d \in D} \left(y^{(d)} - w_0 - \sum_{i=1}^n w_i x_i^{(d)} \right)^2 \quad \text{subject to} \quad \sum_{i=1}^n |w_i| \leq t$$

Ridge regression and Lasso

β 's are the weights
in this figure

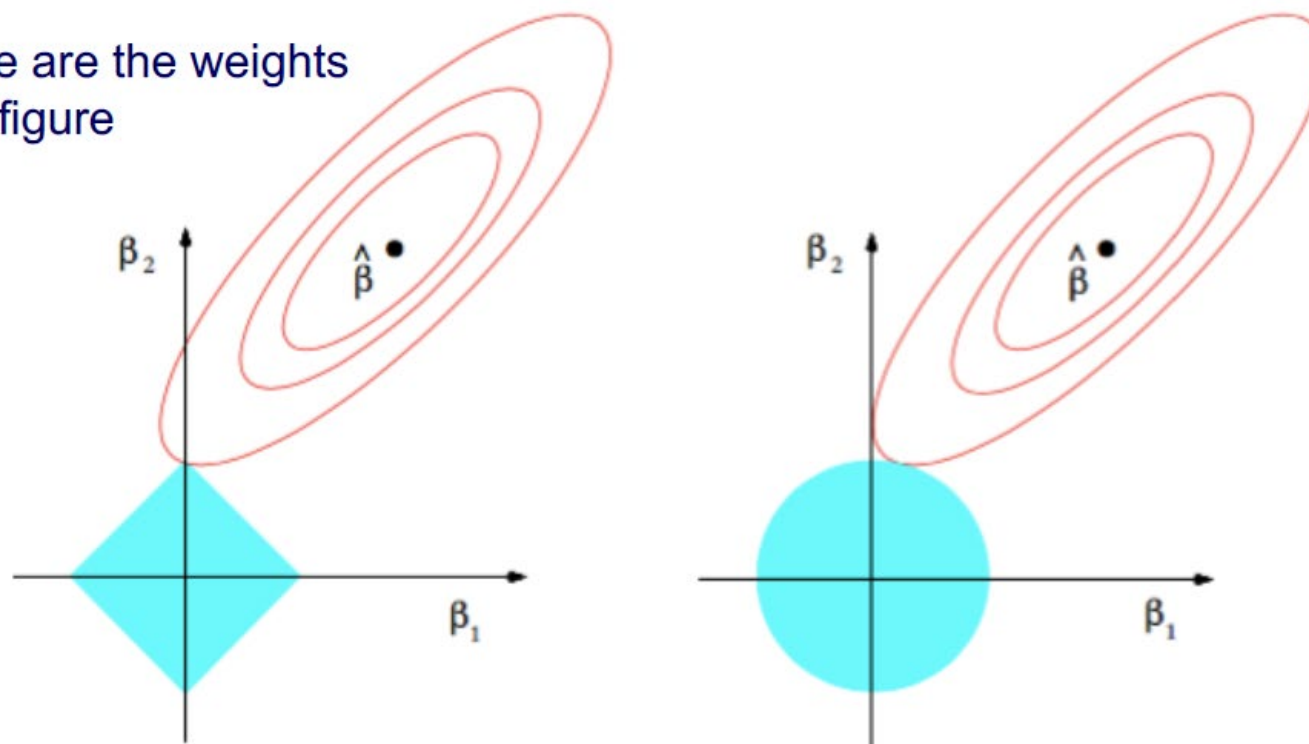


FIGURE 3.11. Estimation picture for the lasso (left) and ridge regression (right). Shown are contours of the error and constraint functions. The solid blue areas are the constraint regions $|\beta_1| + |\beta_2| \leq t$ and $\beta_1^2 + \beta_2^2 \leq t^2$, respectively, while the red ellipses are the contours of the least squares error function.

Feature selection via shrinkage

- Lasso (L1) tends to make many weights 0, inherently performing feature selection
- Ridge regression (L2) shrinks weights but isn't as biased towards selecting features
- L1 and L2 penalties can be used with other learning methods (logistic regression, neural nets, SVMs, etc.)
- Both can help avoid overfitting by reducing variance
- There are many variants with somewhat different biases
 - elastic net: includes L1 and L2 penalties

As an optimization problem, binary class ℓ_2 penalized logistic regression minimizes the following cost function:

$$\text{l1_ratio} = 0 \quad \min_{w,c} \frac{1}{2} w^T w + C \sum_{i=1}^n \log(\exp(-y_i(X_i^T w + c)) + 1).$$

Similarly, ℓ_1 regularized logistic regression solves the following optimization problem:

$$\text{l1_ratio} = 1 \quad \min_{w,c} \|w\|_1 + C \sum_{i=1}^n \log(\exp(-y_i(X_i^T w + c)) + 1).$$

Elastic-Net regularization is a combination of ℓ_1 and ℓ_2 , and minimizes the following cost function:

$$0 \leq \text{l1_ratio} \leq 1 \quad \min_{w,c} \frac{1-\rho}{2} w^T w + \rho \|w\|_1 + C \sum_{i=1}^n \log(\exp(-y_i(X_i^T w + c)) + 1),$$

`l1_ratio` : float or None, optional (default=None)

The Elastic-Net mixing parameter, with `0 <= l1_ratio <= 1`. Only used if `penalty='elasticnet'`. Setting `l1_ratio=0` is equivalent to using `penalty='l2'`, while setting `l1_ratio=1` is equivalent to using `penalty='l1'`. For `0 < l1_ratio < 1`, the penalty is a combination of L1 and L2.

Embedded approach

Logistic regression

```
[45] lr = LogisticRegression(penalty = 'l1', C=0.1, solver='liblinear')  
lr.fit(X_train, y_train)
```

▼ LogisticRegression
LogisticRegression(C=0.1, penalty='l1', solver='liblinear')

Embedded approach

Logistic regression

```
[46] lr.coef_[0]
```

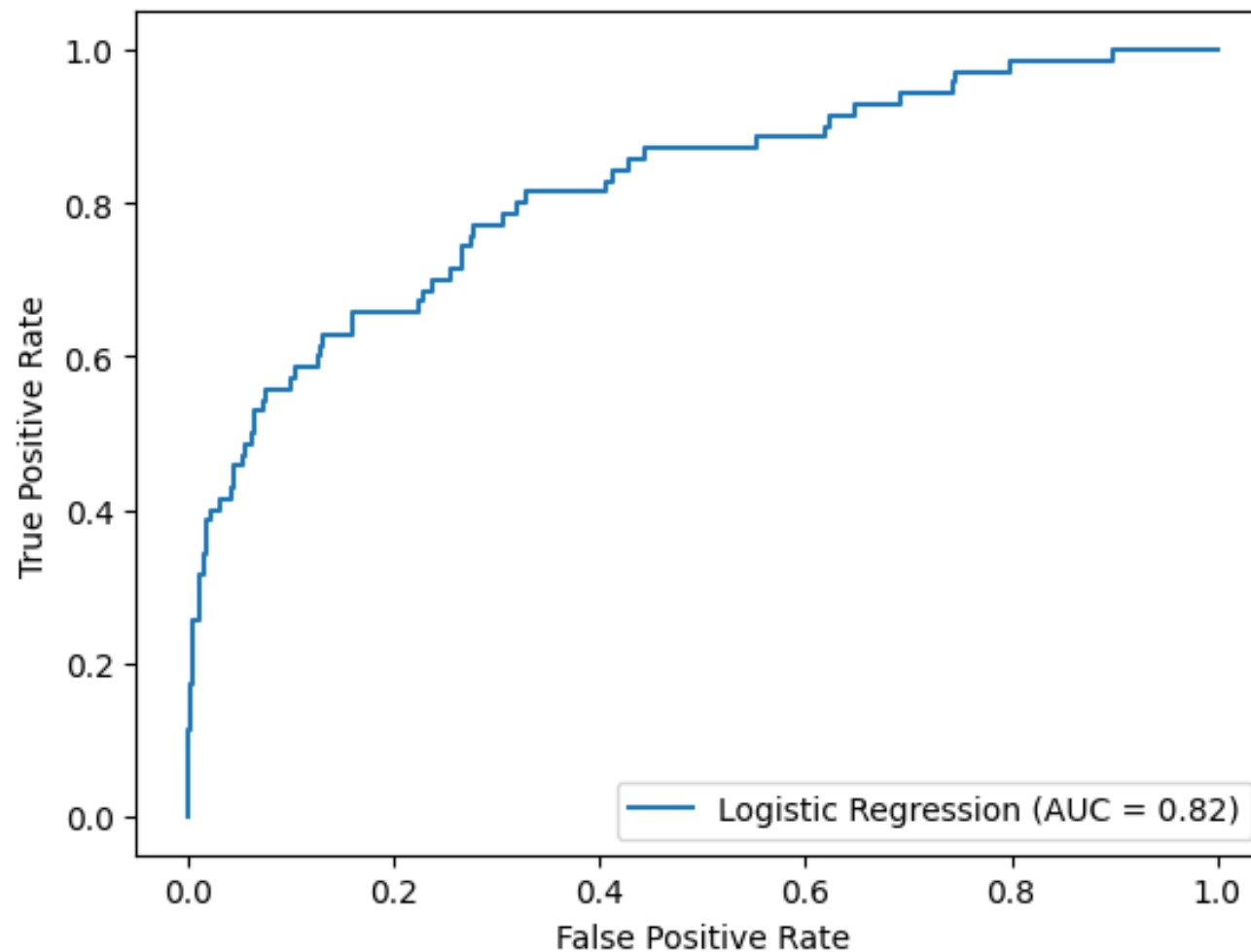
```
array([ 4.39645027e-01,  0.00000000e+00,  0.00000000e+00,  3.64882336e-01,
        0.00000000e+00,  0.00000000e+00,  0.00000000e+00,  0.00000000e+00,
        0.00000000e+00,  0.00000000e+00,  0.00000000e+00,  0.00000000e+00,
        0.00000000e+00,  0.00000000e+00,  0.00000000e+00,  5.38658322e-01,
        1.08934507e+00, -1.10260038e-02, -3.53481268e-04,  3.35990896e-02,
        0.00000000e+00, -2.10427843e-01,  2.69803017e-03, -2.26244263e-01,
        0.00000000e+00, -1.45808259e-01, -1.18404056e-04,  7.08198896e-06,
        1.36897240e-01,  0.00000000e+00,  1.39256563e-01, -1.07618808e-01,
        -1.17913518e-01, -1.94316308e-02, -3.42778865e-02, -7.65126263e-02,
        3.73647156e-02, -1.03708133e-01,  1.20789097e-01, -6.54389495e-02])
```

```
▶ X_train.columns[lr.coef_[0]>0.001]
```

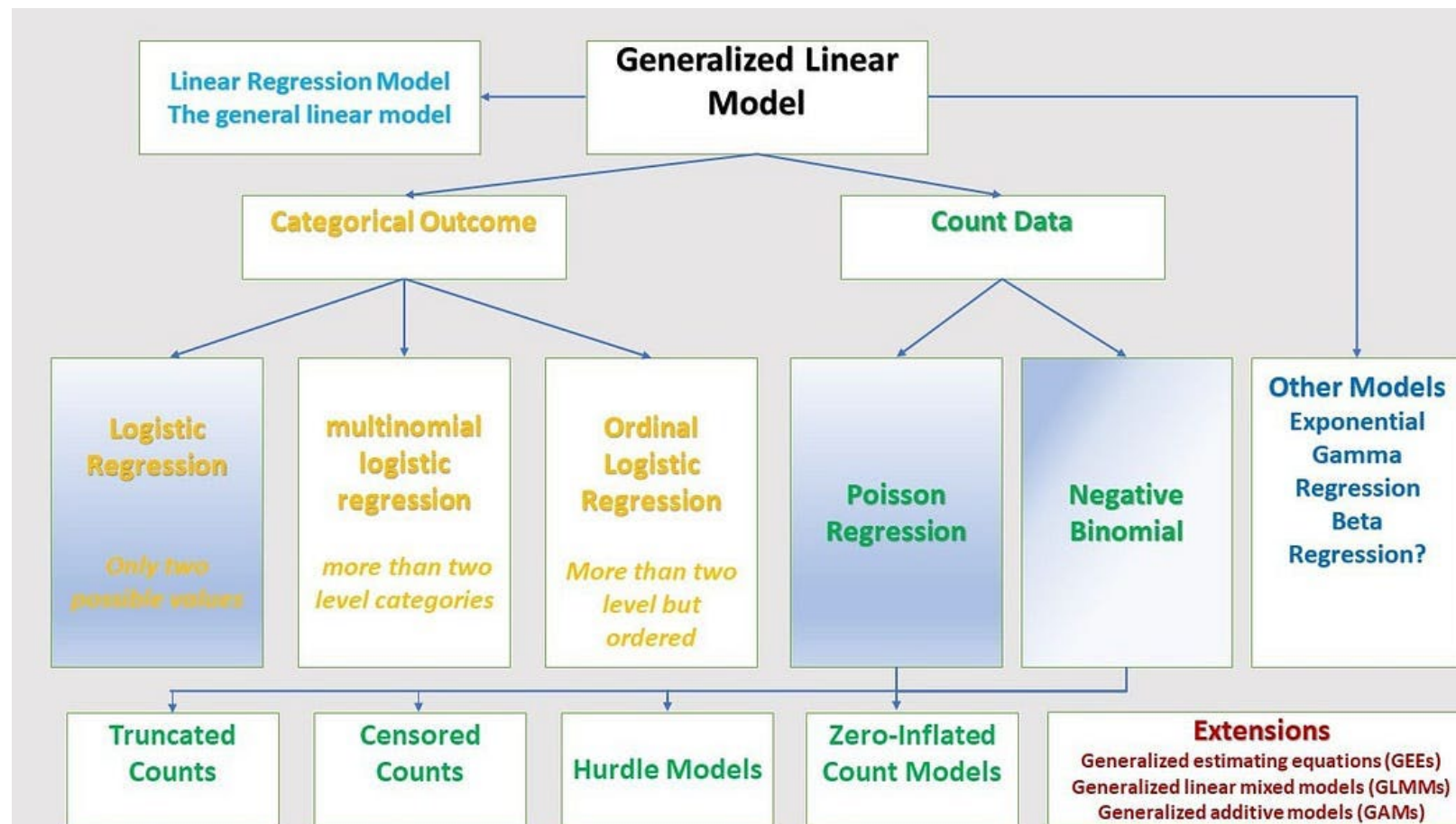
```
Index(['BusinessTravel_Travel_Frequently', 'Department_Sales',
       'MaritalStatus_Single', 'OverTime_Yes', 'DistanceFromHome',
       'HourlyRate', 'NumCompaniesWorked', 'PerformanceRating',
       'YearsAtCompany', 'YearsSinceLastPromotion'],
      dtype='object')
```

Embedded approach

Logistic regression



Generalized linear model





Poisson Regression



Poisson regression

So far we have looked at 2 kinds of regression

- Linear regression (simple, multiple)
 - Continuous response, normally distributed with constant variance
 - Mean a linear function of the covariates
- “Logistic regression”
 - Response (number of successes in n trials) has a binomial distribution $\text{Bin}(n, p)$
 - Mean is np where log-odds of p is a linear function of the covariates

Poisson regression (cont)

- Now we consider “Poisson regression”
 - Response is a count, assumed to have a Poisson distribution with (positive) mean μ
 - Assume that $\log \mu$ is a linear function of the covariates
 - Alternatively,
$$\mu = \exp(\text{linear function of covariates})$$
- Poisson is a standard distribution when response is a count.

Poisson distribution

$$\Pr(Y = y) = \frac{e^{-\mu} \mu^y}{y!}$$

Count Y can have values $0, 1, 2, \dots$

(thus, mean μ must be positive)

The Poisson Regression Model

- The response Y with covariates $x_1, \dots, x_k$ has a Poisson distribution, with mean μ

- Mean μ is related to the covariates by

$$\log(\mu) = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k$$

- As for logistic regression, the parameters are estimated by maximum likelihood

Interpretation of β 's

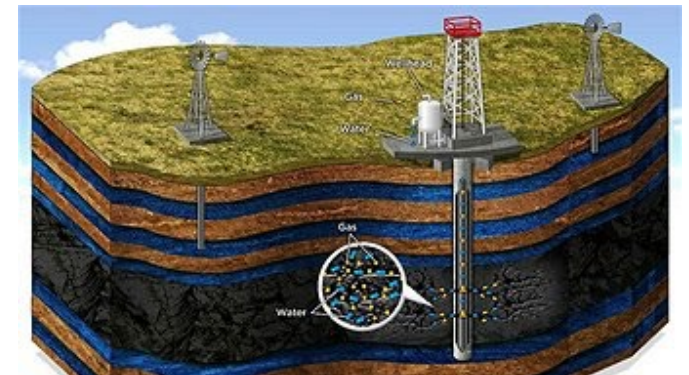
- If $\beta_j > 0$, mean *increases* with x_j
- If $\beta_j < 0$, mean *decreases* with x_j
- Unit increase in x_j changes the mean by a factor of $\exp(\beta_j)$
(like the odds in logistic regression)

Estimation of β 's

- To estimate the β 's , we use the method of *maximum likelihood*, as in logistic regression
- Basic idea:
 - Using the *Poisson* distribution, we can work out the probability of getting any particular set of responses y .
 - In particular, we can work out the probability of getting the data we actually observed – this is the likelihood
 - Choose β 's to maximise this probability (or, equivalently, the log-likelihood)

Example: Mining accident data

- This example features the number of accidents per mine in a 3 month period in 44 coal mines in West Virginia. The variables are
 - COUNT: the number of accidents (response)
 - INB: inner burden thickness
 - EXTRP: percentage of coal extracted from mine
 - AHS: the average height of the coal seam in the mine
 - AGE: the age of the mine





Survival analysis

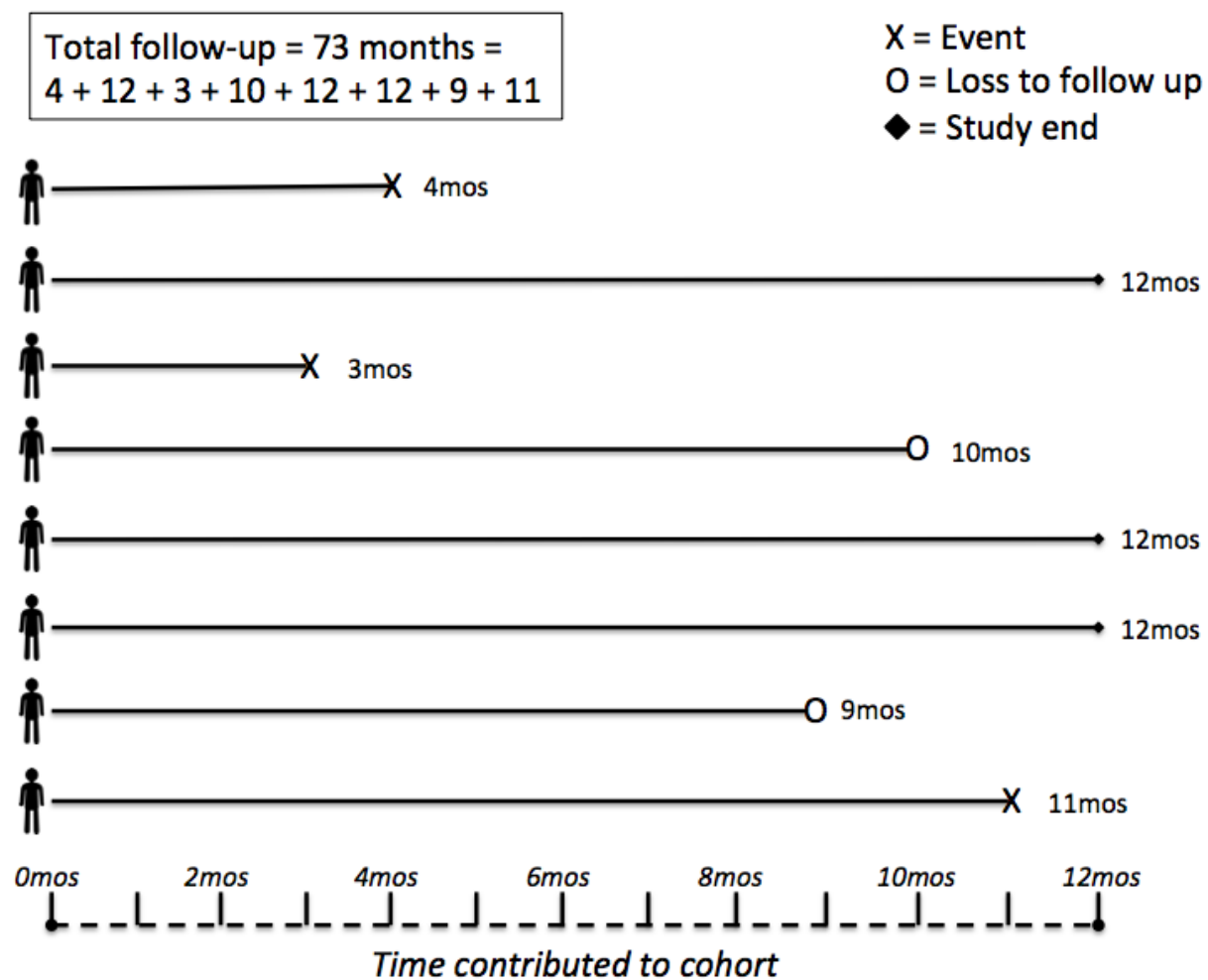


Logistic regression vs time

- In logistic regression, we were interested in studying how risk factors were associated with presence or absence of disease.
- Sometimes, we are interested in how a risk factor or treatment affects time to disease or some other event.
- In these cases, logistic regression is not appropriate.

Survival analysis

- Survival analysis is used to analyze data in which the time until the event is of interest.
- The response is often referred to as a failure time, survival time, or event time.



Example:

- Time until tumor recurrence
- Time until a machine part fails
- Time until mobile phone recharge
- Time until the next credit card usage

The survival time response

- Usually continuous
- May be incompletely determined for some subjects
 - For some subjects we may know that their survival time was at least equal to some time t .
 - Whereas, for other subjects, we will know their exact time of event.
- Incompletely observed responses are **censored**
- Is always ≥ 0 .

Analysis issue

- If there is no censoring, standard linear regression procedures could be used.
- However, these may be inadequate because
 - Time to event is restricted to be positive and has a skewed distribution.
 - The probability of surviving past a certain point in time may be of more interest than the expected time of event.
 - The hazard function, used for regression in survival analysis, can lend more insight into the failure mechanism than linear regression.

Censoring

- Censoring is present when we have some information about a subject's event time, but we don't know the exact event time.
- For the analysis methods we will discuss to be valid, censoring mechanism must be independent of the survival mechanism.

Reasons censoring might occur

- A subject does not experience the event before the study ends
- A person is lost to follow-up during the study period
- A person withdraws from the study

These are all examples of **right-censoring**.

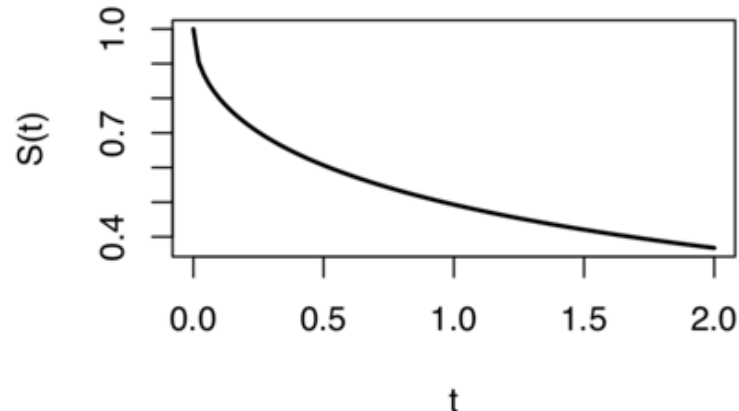
Terminology and notation

- T denotes the response variable, $T \geq 0$.
- The survival function is

$$S(t) = \Pr(T > t) = 1 - F(t).$$

$S(t)$: survival functional

$F(t)$: cumulative distribution of failure events



Survival function

- The survival function gives the probability that a subject will survive past time t .
- As t ranges from 0 to ∞ , the survival function has the following properties
 - It is non-increasing
 - At time $t=0$, $S(t) = 1$. In other words, the probability of surviving past time 0 is 1.
 - At time $t=\infty$, $S(t)=S(\infty)=0$. As time goes to infinity, the survival curve goes to 0.
- In theory, the survival function is smooth.
- In practice, we observe events on a discrete time scale (days, weeks, etc.).

- The hazard function, $h(t)$, is the instantaneous rate at which events occur, given no previous events.

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t < T \leq t + \Delta t | T > t)}{\Delta t} = \frac{f(t)}{S(t)}.$$

- The cumulative hazard describes the accumulated risk up to time t , $H(t) = \int_0^t h(u) du$.

If we know any one of the functions $S(t)$, $H(t)$, or $h(t)$, we can derive the other two functions.

$$h(t) = -\frac{\partial \log(S(t))}{\partial t}$$

$$H(t) = -\log(S(t))$$

$$S(t) = \exp(-H(t))$$

Survival data

How do we record and represent survival data with censoring?

- T_i denotes the response for the i th subject.
- Let C_i denote the censoring time for the i th subject
- Let δ_i denote the event indicator

$$\delta_i = \begin{cases} 1 & \text{if the event was observed } (T_i \leq C_i) \\ 0 & \text{if the response was censored } (T_i > C_i). \end{cases}$$

- The observed response is $Y_i = \min(T_i, C_i)$.

Example

	T_i	C_i	Y_i	δ_i
	80	100	80	1
	40	80	40	1
	74+	74	74	0
	85+	85	85	0
	40	95	40	1

Termination of study

Estimating $S(t)$ and $H(t)$

If we are assuming that every subject follows the same survival function (no covariates or other individual differences), we can easily estimate $S(t)$.

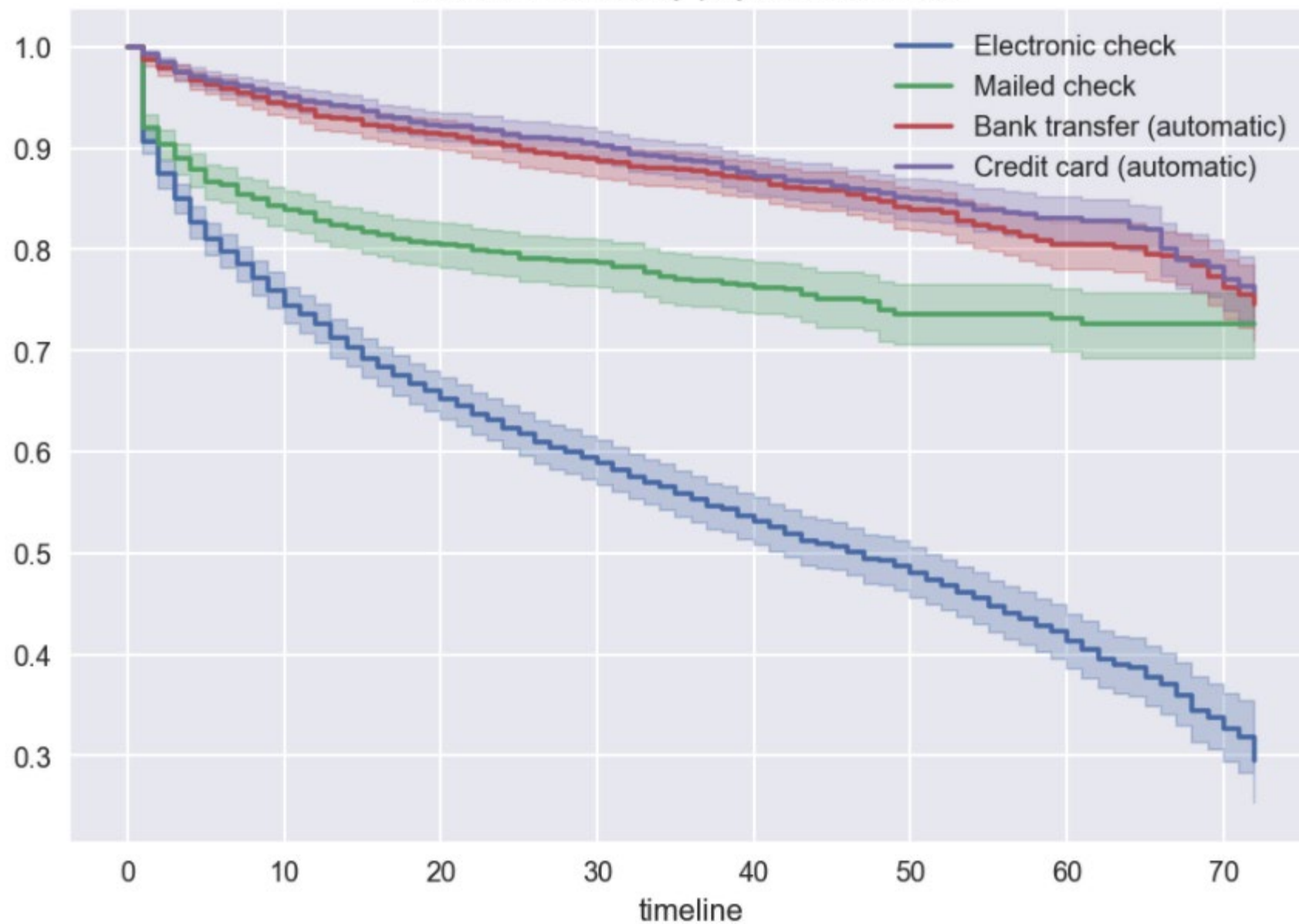
- We can use nonparametric estimators like the Kaplan-Meier estimator
- We can estimate the survival distribution by making parametric assumptions
 - exponential
 - Weibull
 - Gamma
 - log-normal

Non-parametric estimation of S

- When no event times are censored, a non-parametric estimator of $S(T)$ is $1 - F_n(t)$, where $F_n(t)$ is the empirical cumulative distribution function.
- When some observations are censored, we can estimate $S(t)$ using the Kaplan-Meier product-limit estimator.

t	No. subjects at risk	Deaths	Censored	Cumulative survival
59	26	1	0	$25/26 = 0.962$
115	25	1	0	$24/25 \times 0.962 = 0.923$
156	24	1	0	$23/24 \times 0.923 = 0.885$
268	23	1	0	$22/23 \times 0.885 = 0.846$
329	22	1	0	$21/23 \times 0.846 = 0.808$
353	21	1	0	$20/21 \times 0.808 = 0.769$
365	20	0	1	$20/20 \times 0.769 = 0.769$
377	19	0	1	$19/19 \times 0.769 = 0.769$
421	18	0	1	$18/18 \times 0.769 = 0.769$
431	17	1	0	$16/17 \times 0.769 = 0.688$
⋮				⋮
⋮				⋮

Survival curves by payment methods





End of Lecture 4

